Supplementary Information

Bayesian Neural Surrogates Unify Uncertainty Propagation,
Sensitivity Attribution and Calibration for Coupled
Multi-Physics Reactor Analysis

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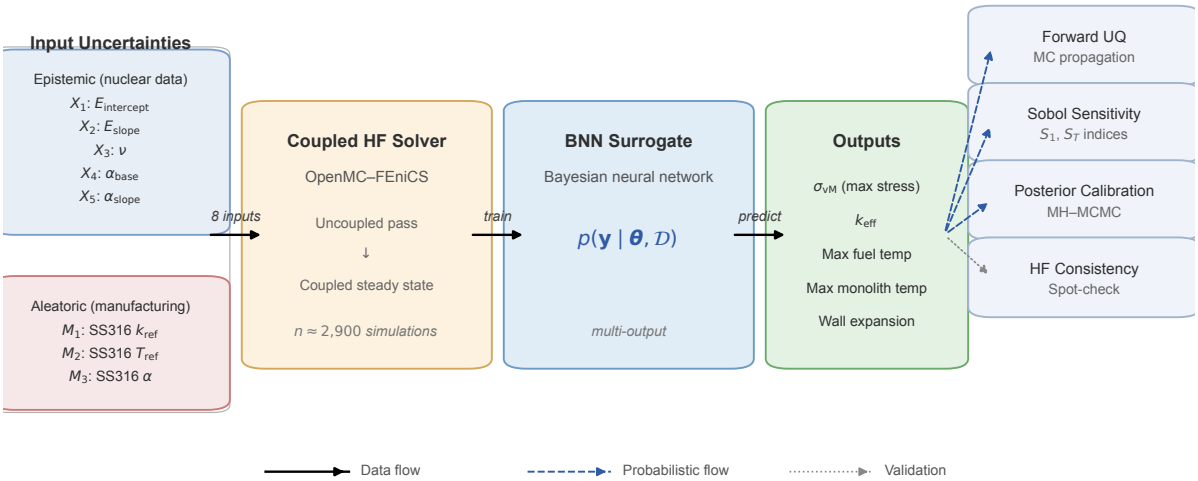


Figure 1: **Fig. S1 | BNN surrogate workflow.** Schematic of the end-to-end probabilistic analysis framework. The BNN surrogate is trained on coupled OpenMC–FEniCS data and produces a posterior predictive distribution. This single distribution serves as the shared probabilistic layer for forward uncertainty propagation, Sobol sensitivity attribution, and MCMC posterior calibration, without retraining between analyses.

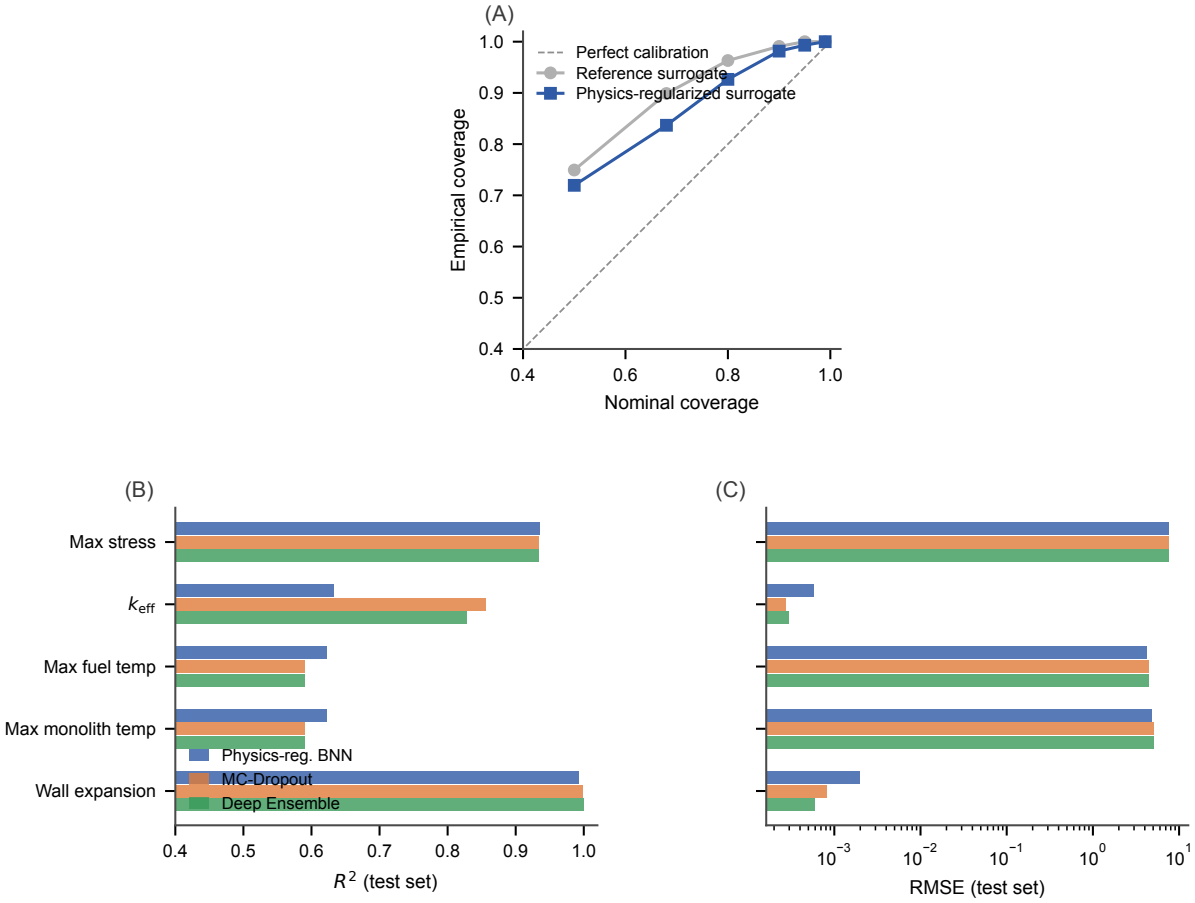
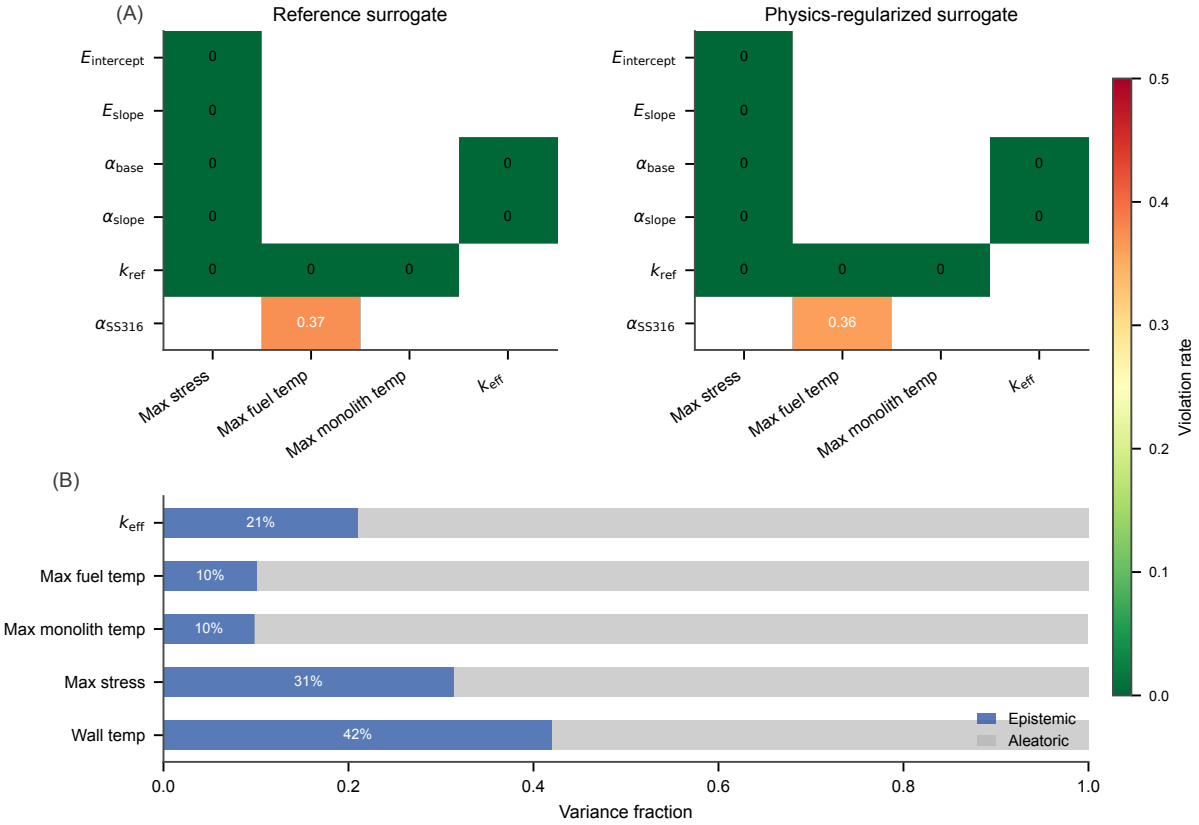


Figure 2: **Fig. S2 | Model validation: parity and residual analysis.** Predicted versus true values and residual distributions for primary outputs across all four BNN variants (435 test samples). Shaded regions show the 90% predictive interval.



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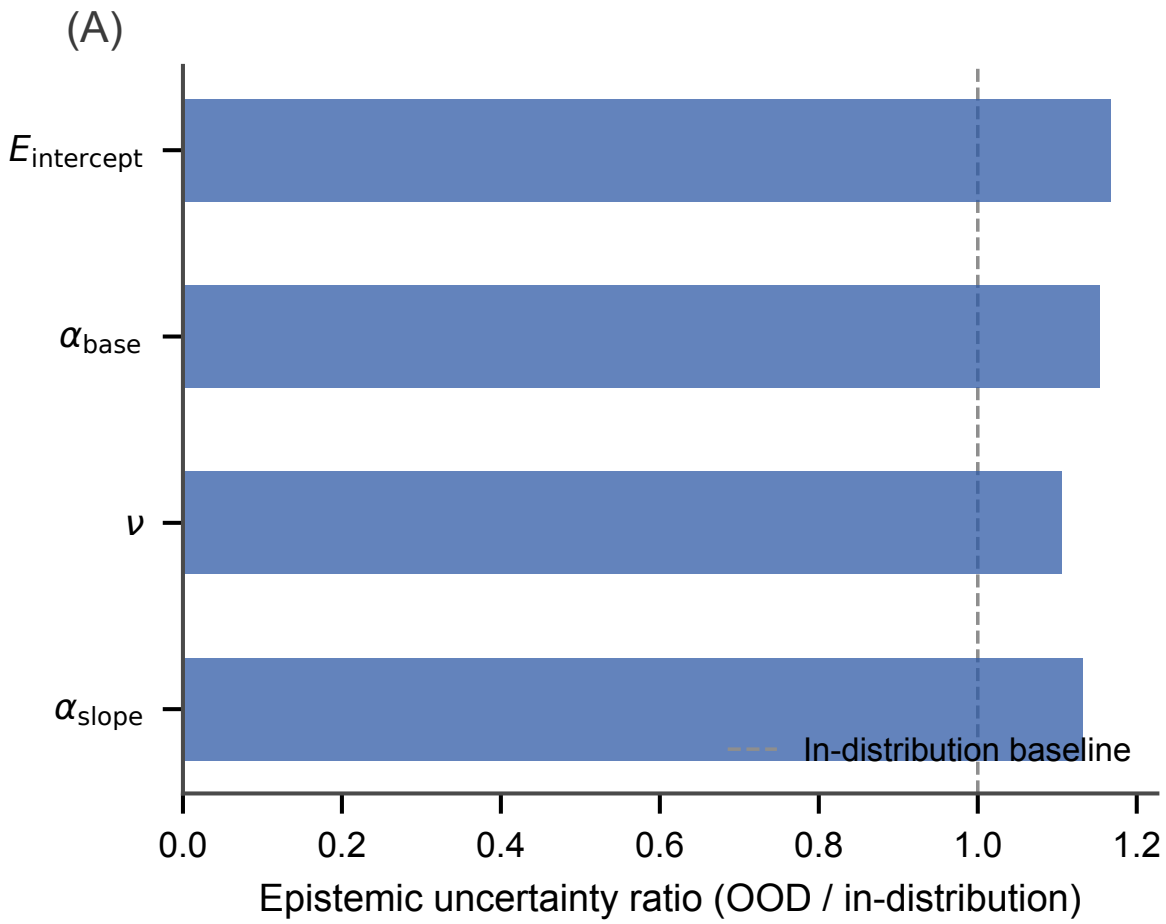

Figure 4: **Fig. S4 | Out-of-distribution (OOD) coverage and epistemic uncertainty.** 90% predictive interval coverage for in-distribution versus four feature-tail subsets. Stress retains coverage above 0.75 across all tails. k_{eff} coverage degrades substantially on the α_{base} tail (R^2 drops to approximately 0.29), indicating that the Sobol and posterior conclusions for k_{eff} are valid only within the training range of this dominant parameter. Elevated epistemic ratios on the α_{base} tail confirm that the BNN correctly signals increased model uncertainty outside the training domain.

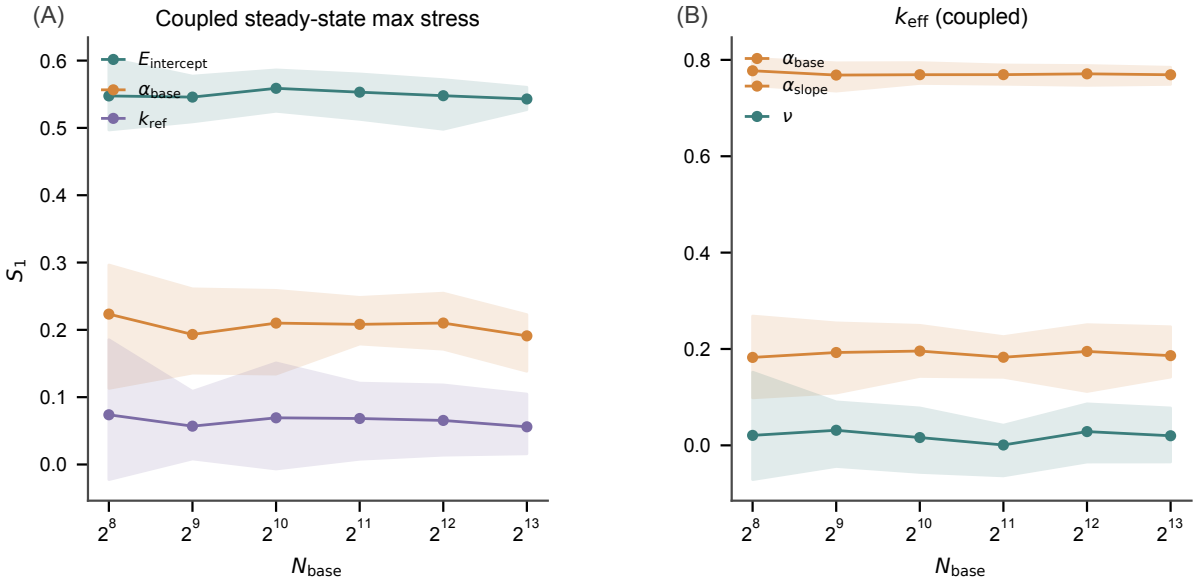

Figure 5: **Fig. S5 | Sobol index convergence with sample size.** First-order Sobol index S_1 as a function of sample size N_S (256 to 4096) for stress and k_{eff}, with 90% CI bands across 50 replications. Indices stabilise by $N_S = 2048$, confirming convergence at the chosen $N_S = 4096$.

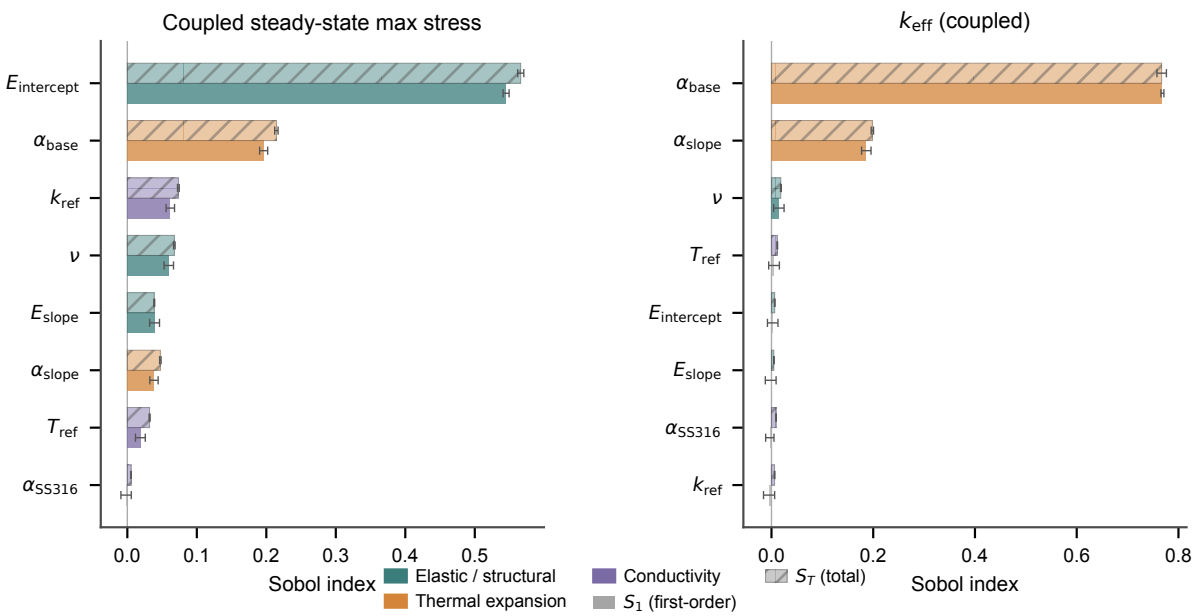

Figure 6: **Fig. S6 | Detailed Sobol sensitivity analysis.** First-order and total-order Sobol indices for all eight material parameters across all five primary outputs, showing the dominance of $E_{\text{intercept}}$ for stress and α_{base} for k_{eff}.

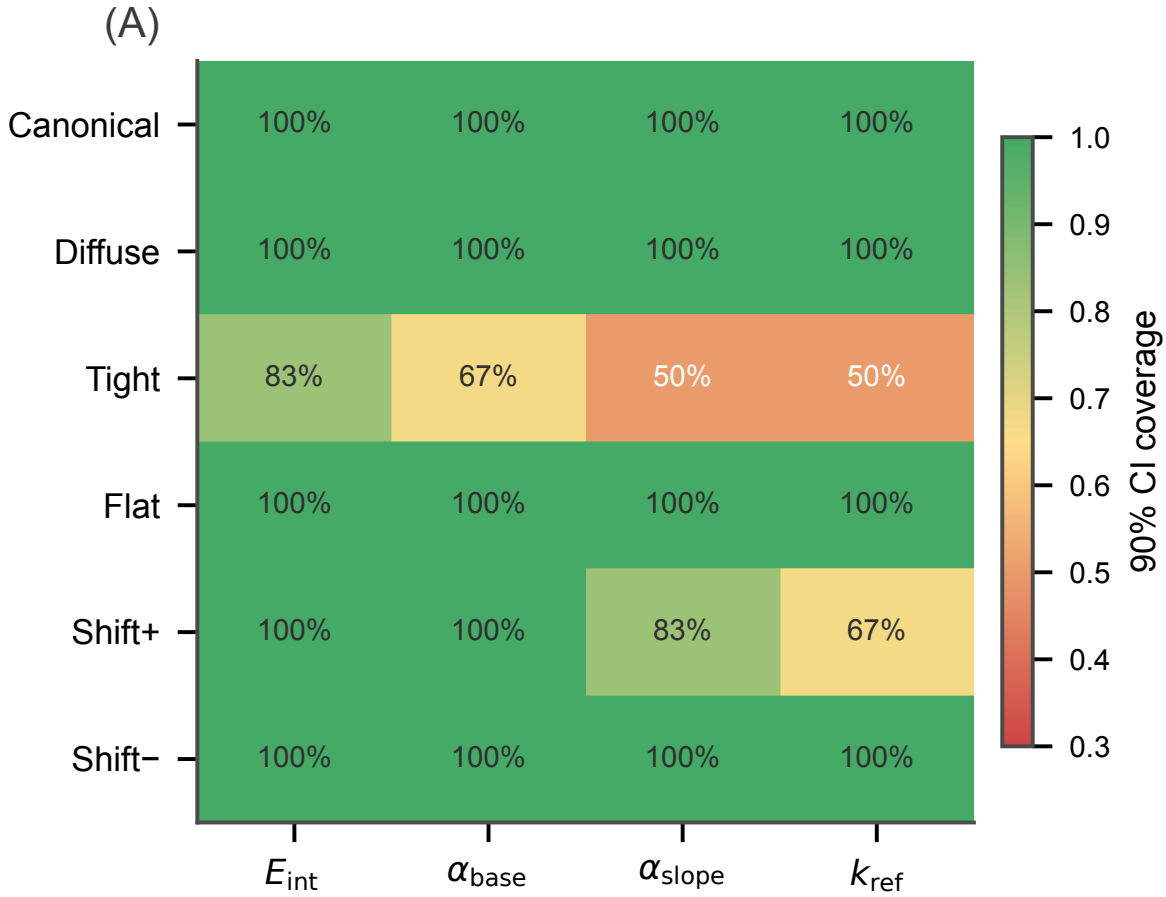

Figure 7: **Fig. S7 | Prior sensitivity analysis.** Posterior marginal distributions for the four calibrated parameters under six prior configurations: canonical, diffuse ($2\times$), tight ($0.5\times$), flat (uniform), shift$_+$ ($\mu + 0.5\sigma$), shift$_-$ ($\mu - 0.5\sigma$). The tight prior over-constrains the posterior, reducing coverage to 50–83% (Table S10).

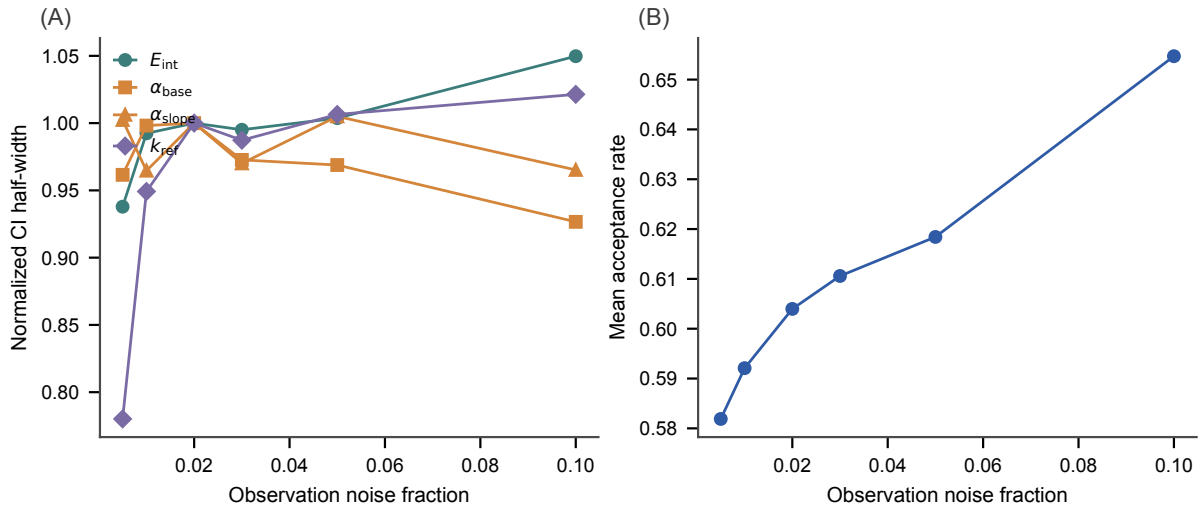

Figure 8: **Fig. S8 | Observation noise sensitivity.** Posterior width and acceptance rate as a function of assumed observation noise fraction (0.5%–10%). Posterior standard deviations increase monotonically; acceptance rates increase appropriately as the likelihood broadens (Table S11).

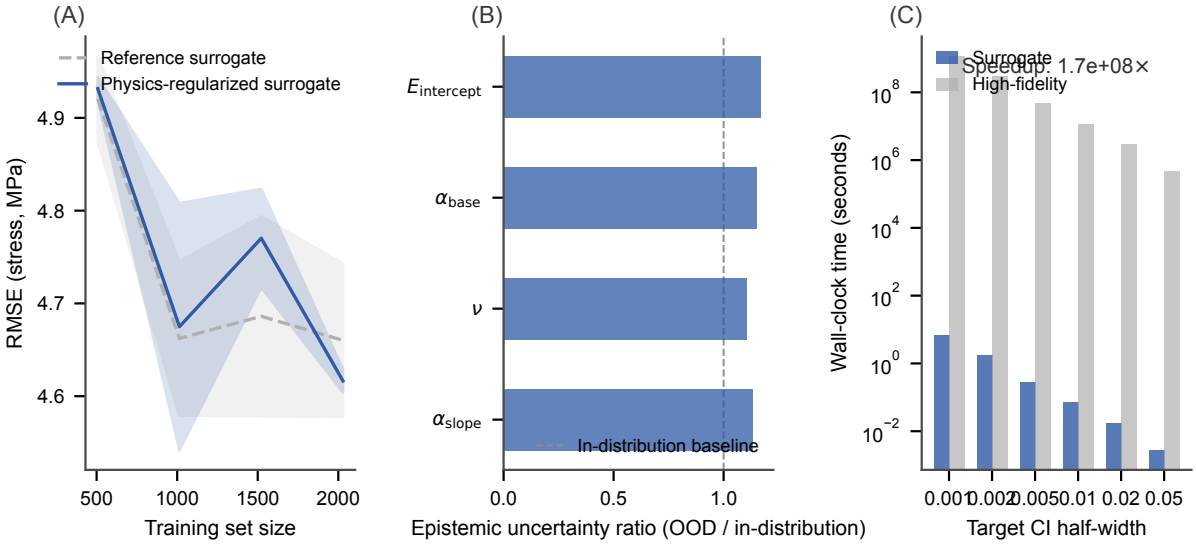

Figure 9: **Fig. S9 | Data efficiency and computational cost.** Test RMSE and R^2 as a function of training set size for the physics-regularized BNN. Near-saturated performance is achieved at 25% data utilisation (507 training samples; Table S12).

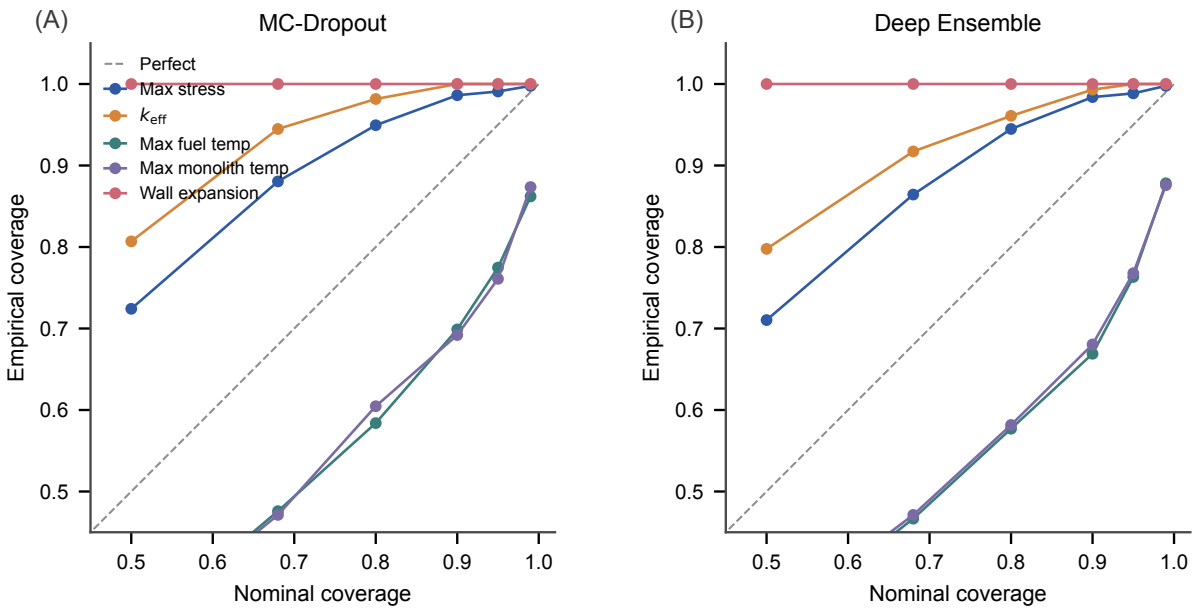

Figure 10: **Fig. S10 | External baseline calibration comparison.** Calibration curves comparing BNN, MC-Dropout and deep ensemble across primary outputs. The BNN achieves substantially better probabilistic calibration (ECE 0.027 vs. 0.18–0.19) on temperature outputs.

2 Supplementary Tables

Table 1: **Table S1 | Material parameters: definitions, nominal values and priors.**

Parameter	Symbol	Nominal value	Unit	Prior range ($\pm$10%)
Young's mod. intercept	$E_{\text{intercept}}$	2×10^{11}	Pa	$[1.8, 2.2] \times 10^{11}$
Young's mod. slope	E_{slope}	-7×10^{7}	Pa/K	$[-7.7, -6.3] \times 10^{7}$
Poisson's ratio	ν	0.31	—	[0.279, 0.341]
CTE base	α_{base}	1×10^{-5}	K^{-1}	$[0.9, 1.1] \times 10^{-5}$
CTE slope	α_{slope}	5×10^{-9}	K^{-2}	$[4.5, 5.5] \times 10^{-9}$
Ref. conductivity	k_{ref}	23.2	W/(m·K)	[20.88, 25.52]
Ref. temperature	T_{ref}	923.15	K	[830.84, 1015.47]
Cond. slope	k_{slope}	1/75	W/(m·K^2)	[0.012, 0.0147]

Table 2: **Table S2 | BNN hyperparameter search space and optimal values.**

Hyperparameter	Search range	Optimal (bnn-phy-mono)
Width	[64, 256]	254
Depth	[2, 8]	2
Learning rate	[10^{-4}, 10^{-2}]	9.81×10^{-4}
Weight decay	[10^{-9}, 10^{-5}]	8.09×10^{-8}
Batch size	{16, 32, 64}	32
Epochs	[200, 500]	399
Gradient clip	[0.1, 2.0]	0.674
w_{data}	[0.1, 5.0]	1.855
Prior σ	[0.1, 2.0]	0.478
KL weight	[10^{-5}, 10^{-2}]	1.93×10^{-4}
w_{mono}	[0.01, 1.0]	0.212
$\rho_{\text{abs,min}}$	[0.01, 0.5]	0.123
mono_topk	[10, 100]	53

Search: Optuna with 200 trials; objective: validation RMSE. Activation: SiLU throughout. Output heads: mean + log-variance (heteroscedastic).

Table 3: **Table S3 | BNN ablation study: four model variants on five primary outputs** (mean $\pm$ s.d., 5 seeds).

Model variant	Stress R^2	Stress PICP	k_{eff} R^2
Reference (ELBO only)	0.936 ± 0.007	0.995 ± 0.002	0.633 ± 0.260
Data-monotone	0.932 ± 0.005	0.998 ± 0.002	0.635 ± 0.260
Physics-regularized	0.935 ± 0.007	0.987 ± 0.007	0.633 ± 0.258
Data+inequality	0.924 ± 0.007	1.000 ± 0.001	0.617 ± 0.249

Table 4: **Table S4 | Complete Sobol indices (S_1 and S_T) for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI).

Parameter	Stress S_1 [90% CI]	Stress S_T [90% CI]
E_{slope}	0.037 [−0.046, 0.100]	0.039 [0.038, 0.039]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	0.566 [0.562, 0.571]
ν	0.063 [0.034, 0.097]	0.068 [0.067, 0.069]
α_{base}	0.210 [0.171, 0.254]	0.214 [0.212, 0.217]
α_{slope}	0.038 [−0.013, 0.091]	0.047 [0.046, 0.049]
T_{ref}	0.019 [−0.025, 0.055]	0.032 [0.031, 0.033]
k_{ref}	0.065 [0.014, 0.118]	0.073 [0.072, 0.075]
SS316_α	−0.004 [−0.068, 0.034]	0.005 [0.005, 0.006]
Parameter	**k_{eff} S_1 [90% CI]**	**k_{eff} S_T [90% CI]**
E_{slope}	0.010 [−0.052, 0.062]	0.005 [0.004, 0.005]
$E_{\text{intercept}}$	0.008 [−0.065, 0.071]	0.006 [0.006, 0.007]
ν	0.029 [−0.035, 0.086]	0.018 [0.018, 0.019]
α_{base}	0.771 [0.747, 0.788]	0.767 [0.758, 0.776]
α_{slope}	0.195 [0.112, 0.250]	0.198 [0.196, 0.201]
T_{ref}	0.010 [−0.023, 0.054]	0.011 [0.011, 0.012]
k_{ref}	0.000 [−0.063, 0.048]	0.006 [0.006, 0.007]
SS316_α	0.003 [−0.059, 0.074]	0.009 [0.009, 0.009]

Table 5: **Table S5 | Posterior calibration: 18 benchmark case summary** (physics-regularized BNN).

Case	Category	True stress (MPa)	Accept rate	90%-CI coverage	Post-mean err.
0	low	110.6	0.619	3/4	8.3
1	low	118.7	0.612	4/4	24.6
2	low	110.7	0.620	2/4	9.0
3	low	103.7	0.618	4/4	8.8
4	low	114.5	0.621	4/4	3.1
5	low	102.0	0.616	3/4	23.4
6	near	127.6	0.609	4/4	20.3
7	near	127.9	0.586	4/4	26.6
8	near	127.0	0.611	4/4	5.9
9	near	123.5	0.613	4/4	3.5
10	near	128.9	0.609	3/4	4.5
11	near	129.3	0.610	4/4	3.8
12	high	169.5	0.592	4/4	0.2
13	high	176.0	0.591	4/4	12.1
14	high	193.4	0.598	3/4	13.7
15	high	163.4	0.591	4/4	3.6
16	high	142.8	0.582	4/4	10.5
17	high	197.6	0.583	4/4	5.4
Overall			0.58–0.62	66/72 = 0.917	MAE = 10.4

Table 6: **Table S6 | Computational cost breakdown.**

Operation	Time	Platform
Coupled HF solve (1 case)	2266 s	EPYC 9654 / 96 cores
BNN single draw ($M = 50$)	1.29×10^{-2} s	GPU
BNN batched (batch=1024, per sample)	1.34×10^{-5} s	GPU
20,000-sample forward UQ	$\sim$0.3 s	GPU
Sobol (50 reps $\times$ 40,960 evals)	$\sim$minutes	GPU
18-case MCMC (8000 iter $\times$ 4 chains)	$\sim$minutes	GPU
Speedup (single draw vs. HF)	1.76×10^5	
Speedup (batched vs. HF)	1.69×10^8	

Table 7: **Table S7 | Physics consistency: monotonicity violation rates** (435 test samples, discrete $\pm\delta$ perturbation test).

Physics pair (input $\rightarrow$ output, sign)	Ref. BNN	Data-mono	Phy-reg	Data+ineq
$E_{\text{intercept}} \rightarrow$ Max stress (+)	0%	0%	0%	0%
$\alpha_{\text{base}} \rightarrow$ Max stress (+)	0%	0%	0%	0%
$\alpha_{\text{base}} \rightarrow k_{\text{eff}}$ ($-$)	0%	0%	0%	0%
$k_{\text{ref}} \rightarrow$ Max stress ($-$)	0%	0%	0%	1.1%
SS316_$\alpha \rightarrow$ Fuel temp ($-$)	36.3%	39.5%	35.6%	53.6%

Inequality constraints (max $\geq$ average temp, stress ≥ 0): 0% violations for all four BNN variants.

Table 8: **Table S8 | Epistemic vs. aleatoric uncertainty decomposition on test set.**

	Ref. BNN		Phy-reg BNN	
Output	Epi%	Ale%	Epi%	Ale%
k_{eff}	12.0	88.0	10.2	89.8
Max fuel temp	12.0	88.0	10.1	89.9
Max monolith temp	11.2	88.8	10.1	89.9
Max stress	27.7	72.3	29.6	70.4
Wall expansion	34.1	65.9	41.7	58.3

Table 9: **Table S9 | External baseline: full comparison of scoring rules.**

Model	Output	RMSE	R^2	CRPS	NLL	ECE
MC-Dropout	k_{eff}	0.0003	0.856	0.0002	-6.502	0.152
	Fuel temp	4.497	0.590	2.624	3.352	0.183
	Monolith T	5.061	0.590	2.951	3.472	0.182
	Max stress	7.659	0.934	4.519	3.592	0.118
	Wall exp.	0.0008	0.999	0.002	-3.831	0.197
Deep Ensemble	k_{eff}	0.0003	0.828	0.0002	-6.476	0.142
	Fuel temp	4.495	0.590	2.634	3.326	0.190
	Monolith T	5.061	0.590	2.962	3.439	0.185
	Max stress	7.644	0.934	4.502	3.586	0.112
	Wall exp.	0.0006	0.999	0.002	-3.841	0.197

Table 10: **Table S10 | Prior sensitivity** (6 configurations, physics-regularized BNN, case 06, near-threshold).

Prior variant	Coverage (4 params)	Mean KL vs. canonical	Accept rate
Canonical	100%	0.00 (reference)	0.61
Diffuse ($2\times$)	100%	0.15–0.22	0.63
Tight ($0.5\times$)	50–83%	0.64–1.28	0.57
Flat (uniform)	100%	0.24–0.38	0.64
Shift ($+0.5\sigma$)	67–100%	0.06–0.36	0.60
Shift (-0.5σ)	83–100%	0.08–0.28	0.61

Table 11: **Table S11 | Noise sensitivity** (noise fraction 0.5%–10%, physics-regularized BNN, case 06).

Noise fraction	$E_{\text{intercept}}$ post. std	Accept rate	Coverage
0.5%	1.26×10^{10}	0.58	100%
1.0%	1.28×10^{10}	0.59	100%
2.0% (default)	1.32×10^{10}	0.61	100%
5.0%	1.36×10^{10}	0.63	100%
10.0%	1.40×10^{10}	0.65	83%

Table 12: **Table S12 | Data efficiency** (2 seeds per fraction).

Train fraction	N_{train}	RMSE (mean $\pm$ s.d.)	R^2 (mean $\pm$ s.d.)
25%	507	4.93 ± 0.04	0.768 ± 0.002
50%	1015	4.66 ± 0.11	0.782 ± 0.005
75%	1522	4.68 ± 0.01	0.784 ± 0.002
100%	2029	4.62 ± 0.08	0.783 ± 0.005

3 Supplementary Notes

3.1 Note A: BNN training details

A.1. Evidence lower bound (ELBO) derivation

The BNN posterior over weights $\mathbf{w}$ is approximated by a mean-field variational distribution $q(\mathbf{w}) = \prod_i \mathcal{N}(w_i \mid \mu_i, \sigma_i^2)$. The variational parameters are optimised by minimising the negative evidence lower bound:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathrm{KL}\big[q(\mathbf{w}) \,\|\, p(\mathbf{w})\big] - \mathbb{E}_{q(\mathbf{w})}\big[\log p(\mathcal{D} \mid \mathbf{w})\big] \tag{1}$$

where $p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$ is the isotropic Gaussian prior. The KL divergence admits a closed form; the data-fit term is approximated by a single Monte Carlo sample per mini-batch (Bayes-by-Backprop[10]).

The reparameterisation trick enables gradient computation: $w = \mu + \sigma \cdot \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. The variance parameter $\sigma = \log(1 + \exp(\rho))$ ensures positivity, with $\rho_{\text{abs,min}}$ clamping $\rho \geq \rho_{\text{abs,min}}$ to prevent posterior collapse.

A.2. Heteroscedastic observation noise

The network outputs both a mean $\mu(\boldsymbol{\theta})$ and a log-variance $\log \sigma^2_{\text{obs}}(\boldsymbol{\theta})$ per output channel:

$$\log p(y_j \mid \boldsymbol{\theta}, \mathbf{w}) = -\tfrac{1}{2}\big[\log \sigma^2_{\text{obs},j} + (y_j - \mu_j)^2 / \sigma^2_{\text{obs},j}\big] \tag{2}$$

This heteroscedastic formulation allows the BNN to model input-dependent observation noise, separating it from epistemic (weight) uncertainty.

A.3. KL annealing schedule

The KL weight β is annealed from 0 to its target value kl_weight $= 1.93 \times 10^{-4}$ over the first 100 epochs: $\beta(t) = \min(t/100, 1) \cdot$ kl_weight. This prevents the KL term from dominating early training before the data-fit term has found a reasonable region of weight space.

A.4. Physics-prior monotonicity constraints

The physics-regularized BNN adds an auxiliary loss:

$$\mathcal{L}_{\text{mono}} = \frac{1}{K} \sum_{k=1}^{K} \max\big(0,\ -\text{sign} \cdot \partial\mu / \partial\theta_k\big) \tag{3}$$

weighted by $w_{\text{mono}} = 0.212$. Sign constraints are derived from constitutive physics—for example, stress increases with $E_{\text{intercept}}$ because $\sigma = \mathbf{C}(E, \nu) \cdot \boldsymbol{\epsilon}$ and $\mathbf{C}$ is linear in E.

These are physics-prior constraints, not inequality priors. The inequality-constraint variant (tested in ablation, Table S3) is a separate formulation.

A.5. Gradient clipping and initialisation

Gradient norms are clipped at 0.674 per parameter group. Weight means μ are initialised with Xavier uniform; ρ values are initialised uniformly in $[-5, -4]$, corresponding to small initial variance. These

choices stabilise early training when the KL term is annealed in.

3.2 Note B: Sobol sensitivity methodology

B.1. Sobol variance decomposition

For a square-integrable function $f(\theta_1, \ldots, \theta_d)$ with independent inputs, the total variance $V = \mathrm{Var}[f]$ can be decomposed as:

$$V = \sum_i V_i + \sum_{i<j} V_{ij} + \cdots + V_{1,\ldots,d} \tag{4}$$

The first-order Sobol index is $S_i = V_i/V$; the total-order index is $S_{T,i} = 1 - V_{\sim i}/V$.

B.2. Saltelli cross-matrix scheme

Two independent $N \times d$ matrices $\mathbf{A}$ and $\mathbf{B}$ are drawn from the joint prior. For each input i, a cross-matrix $\mathbf{A}_{\mathbf{B}}^{(i)}$ is constructed by replacing the i-th column of $\mathbf{A}$ with the i-th column of $\mathbf{B}$. The model is evaluated at all $N(d+2)$ points.

B.3. Jansen estimator

$$S_{1,i} = 1 - \frac{1}{2N} \sum_{j=1}^{N} \big[f(\mathbf{A}_{\mathbf{B}}^{(i)})_j - f(\mathbf{B})_j \big]^2 / V \tag{5}$$

$$S_{T,i} = \frac{1}{2N} \sum_{j=1}^{N} \big[f(\mathbf{A})_j - f(\mathbf{A}_{\mathbf{B}}^{(i)})_j \big]^2 / V \tag{6}$$

The Jansen estimator is preferred for its lower variance compared to the Sobol estimator, particularly for indices near zero.

B.4. Replication-based confidence intervals

The entire Sobol analysis (sample generation, model evaluation, index estimation) is independently repeated $R = 50$ times. The 90% confidence interval for each index is constructed from the 5th and 95th percentiles of the R replications. A factor is declared a stable contributor only if the lower bound of its 90% CI exceeds zero.

The BNN posterior predictive mean (averaged over $M = 30$ weight samples) serves as the deterministic function $f(\boldsymbol{\theta})$ in the Sobol framework.

3.3 Note C: MCMC posterior calibration details

C.1. Likelihood formulation

$$\log p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \sum_{j=1}^{J} \log \mathcal{N}\big(y_{\text{obs},j} \mid \mu_j(\boldsymbol{\theta}),\ \sigma^2_{\text{obs},j} + \sigma^2_{\text{BNN},j}(\boldsymbol{\theta})\big) \tag{7}$$

where $\mu_j(\boldsymbol{\theta})$ is the BNN posterior predictive mean (averaged over $M = 20$ weight samples), $\sigma^2_{\text{BNN},j}$ is the BNN aleatoric variance, and $\sigma^2_{\text{obs},j}$ is the observation noise variance (2% of the true value). The $J = 5$

primary outputs are: coupled peak global stress, coupled peak fuel temperature, coupled peak monolith temperature, coupled wall expansion, and k_{eff}.

C.2. Proposal distribution

A symmetric Gaussian random-walk proposal is used: $\boldsymbol{\theta}^* = \boldsymbol{\theta}_{\text{current}} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{proposal}})$, where $\boldsymbol{\Sigma}_{\text{proposal}} = \text{diag}(\text{scale}_1^2, \ldots, \text{scale}_4^2)$. The proposal scales were calibrated by short pilot runs of 1000 iterations.

C.3. Chain configuration

For each benchmark case:

- 4 independent chains with dispersed initialisations drawn from the prior
- $N_{\text{total}} = 8000$ iterations per chain
- Burn-in: first 2000 iterations discarded
- Thinning: every 5th post-burn-in sample retained
- Effective posterior: 1200 samples per chain, 4800 total per case

The four calibrated parameters are $E_{\text{intercept}}$, α_{base}, α_{slope}, and k_{ref}, selected as the four with the largest Sobol total-order indices across stress and k_{eff}. The remaining four parameters (E_{slope}, ν, T_{ref}, k_{slope}) are fixed at their true values.

C.4. Convergence diagnostics

Convergence is assessed by:

1. Visual inspection of trace plots (Fig. S9 in the main text references)
2. Rank-normalised split-$\hat{R}$ (Vehtari et al. 2021): $\hat{R}_{\text{max}} = 1.017$ across all 72 parameter–case combinations, well below the 1.1 threshold
3. Geyer initial-positive-sequence effective sample size: $\text{ESS}_{\text{min}} = 352$ (all cases)
4. Acceptance rates: range 0.58–0.62, mean 0.605 (Table S5)

C.5. Benchmark case selection

The 18 benchmark cases are drawn from the 435-sample test split:

- 6 low-stress cases (true stress 102–119 MPa)
- 6 near-threshold cases (true stress 123–129 MPa)
- 6 high-stress cases (true stress 143–198 MPa)

The threshold for categorisation is the primary safety threshold of 131 MPa. For each case, the "observation" is the true HF output vector perturbed by 2% multiplicative Gaussian noise.

3.4 Note D: Forward UQ extensions

D.1. Posterior predictive mean vs. full predictive draw

The main-text forward UQ propagates through the BNN posterior predictive mean $y = \mathbb{E}_{\mathbf{w}}[\mu_{\mathbf{w}}(\boldsymbol{\theta})]$, isolating parametric uncertainty from the BNN's own epistemic and aleatoric spread. As a consistency check, full predictive draws (sampling both weight uncertainty and observation noise) produce wider distributions but consistent means and percentile rankings, confirming that the parametric sensitivity structure is robust to the choice of predictive protocol.

D.2. Latin hypercube sampling

Latin hypercube sampling (LHS) is used rather than simple random sampling because LHS ensures more uniform coverage of the input space, reducing sampling variance for a given sample size. With $N = 20{,}000$ samples and $d = 8$ inputs, the LHS design partitions each input dimension into 20,000 equal-probability intervals.

3.5 Note E: Coupled simulation and material model details

E.1. Material constitutive models

The SS316 monolith is the structural component that simultaneously bears thermal stress and transmits heat between fuel and heat pipes. All eight uncertain parameters enter the calculation through the constitutive models:

- **Structural mechanics:** $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$; ν (Poisson's ratio). Stress: $\sigma = \mathbf{C}(E, \nu) \cdot \boldsymbol{\epsilon}$.
- **Thermal deformation:** $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$. Dual role: thermal strain $\epsilon_{\text{th}} = \alpha \cdot \Delta T$ and geometric expansion.
- **Thermal transport:** $k(T) = k_{\text{ref}} + k_{\text{slope}} \cdot (T - T_{\text{ref}})$.

E.2. Picard iteration and convergence

The OpenMC–FEniCS coupling proceeds as:

1. Initialise with the eight material parameters and reference geometry.
2. OpenMC computes k_{eff} and the spatially resolved fission power distribution.
3. FEniCS solves for the temperature field, thermal expansion displacement and stress field.
4. The updated geometry and temperature-dependent material cross-sections are fed back to OpenMC.
5. Steps 2–4 repeat until convergence.

Convergence criterion: maximum temperature difference between consecutive iterations $< 1\,\text{K}$. Under-relaxation factor: 0.1.

E.3. Output definitions

Each HF evaluation produces 15 scalar outputs: 7 decoupled (before coupling convergence) and 8 coupled (after full convergence, including k_{eff}). The posterior likelihood uses 5 primary outputs: k_{eff}, peak fuel

temperature, peak monolith temperature, peak global stress, and wall expansion.

E.4. Dataset generation

$n = 2900$ samples were generated by Latin hypercube sampling over the 8-dimensional prior. The split (seed $= 2026$): training 2029 ($\sim$70%), validation 436 ($\sim$15%), test 435 ($\sim$15%). The fixed split is reused across all experiments for reproducibility.

Supplementary References

9. Zhang T, Chen Y, Wang S, Liu X. Design-oriented multiphysics analysis and surrogate modelling of a heat-pipe-cooled reactor. *Energy*. 2025. (in press)
10. Blundell C, et al. Weight uncertainty in neural networks. ICML 2015; pp. 1613–1622.